

Quadratic Equations (Algebra)

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Part I

Theory

Part II

Problem Set

Chapter 1

JEE Mains Problems

Q1: Let $\alpha, \alpha + 2, \alpha \in \mathbb{Z}$, be the roots of the quadratic equation
$$x(x + 2) + (x + 1)(x + 3) + (x + 2)(x + 4) + \dots + (x + n + 1)(x + n - 1) = 4n$$
 for some $n \in \mathbb{N}$. Then $n + \alpha$ is equal to :
A) 0
B) 1
C) 2
D) 3

{Solution: <https://www.youtube.com/watch?v=AVSUet9a0>}